# A Survey and Comparative Analysis of Image Processing Techniques for Diabetic Retinopathy Detection

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***Abstract*—Diabetic retinopathy (DR) is a progressive retinal complication of diabetes and a leading cause of preventable blindness worldwide. Early detection through automated analysis of retinal fundus photographs has attracted significant research interest over the past two decades. This paper presents a survey and comparative analysis of image processing and machine learning techniques applied to DR detection, covering classical approaches based on morphological operations and thresholding, as well as modern deep learning pipelines. We systematically examine the role of preprocessing techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE), green channel extraction, and Gaussian filtering in enhancing fundus image quality. We then analyze segmentation strategies including Otsu thresholding, morphological operations, and U-Net based approaches for isolating retinal lesions such as microaneurysms, exudates, and hemorrhages. Finally, we discuss classification approaches ranging from support vector machines to convolutional neural networks, summarizing reported performance across commonly used benchmark datasets. The goal of this survey is to consolidate understanding of the image processing foundations underlying DR screening systems and to identify remaining challenges in the field.**

***Index Terms*—diabetic retinopathy, fundus image processing, CLAHE, image segmentation, convolutional neural network, lesion detection, retinal image analysis, survey**

## I. Introduction

Diabetes mellitus affects an estimated 537 million adults globally as of 2021, a number projected to reach 783 million by 2045 [1]. Among its most damaging long-term complications is diabetic retinopathy, a condition in which prolonged hyperglycemia causes progressive damage to the microvasculature of the retina. In its advanced stages, DR leads to vitreous hemorrhage, retinal detachment, and irreversible blindness. Globally, DR is responsible for approximately 4.8% of all cases of blindness, making it one of the most significant causes of preventable vision loss [2].

The current clinical standard for DR screening involves trained ophthalmologists manually examining retinal fundus photographs and grading them on a severity scale. While reliable, this process is slow, expensive, and dependent on specialist availability — a critical bottleneck in countries such as India, where diabetic populations are large but ophthalmic infrastructure in rural areas is limited [3]. Automated image analysis offers a scalable alternative: algorithms can process fundus images rapidly and consistently, flagging cases that require urgent clinical attention.

The retinal fundus image contains a rich set of pathological markers that correspond to DR severity. These include:

- **Microaneurysms (MA):** Small, round, dark red dots caused by weakening of capillary walls; typically the earliest visible sign of DR.
- **Hemorrhages:** Larger, irregular dark blotches resulting from rupture of blood vessels.
- **Hard Exudates (HE):** Bright yellow-white deposits of lipid and protein that leak from damaged vessels.
- **Cotton Wool Spots (CWS):** Soft, fluffy white patches caused by localized nerve fiber layer infarctions; indicative of severe ischemia.
- **Neovascularization:** Abnormal growth of new, fragile blood vessels; a hallmark of proliferative DR.

Each of these features has distinct visual properties in the image domain, making fundus photography well-suited to digital image processing analysis. The challenge lies in the variability of image quality, illumination, patient anatomy, and camera hardware across clinical settings.

This paper surveys and analyzes the spectrum of image processing techniques that have been applied to automated DR detection. Rather than presenting new experimental results, we aim to consolidate the methodological foundations of the field, critically evaluate the trade-offs between different approaches, and discuss the key design decisions involved in building a DR screening pipeline. This analysis is organized around three core stages: image preprocessing and enhancement, lesion segmentation, and classification.

## II. Fundus Image Characteristics and Challenges

A retinal fundus image is a two-dimensional projection of the three-dimensional curved retinal surface, captured using a fundus camera. Standard images are color photographs in RGB space, typically at resolutions ranging from $1500 \times 1500$ to $4288 \times 2848$ pixels depending on the imaging system.

Several properties of fundus images make automated analysis non-trivial:

**Illumination non-uniformity:** The center of the image is typically better illuminated than the periphery due to the optics of the fundus camera. This creates a gradient in pixel intensity across the image, which can interfere with global thresholding and feature extraction.

**Low contrast of pathological regions:** Microaneurysms, in particular, are small ($\leq 125\ \mu m$ in diameter) and can have intensity values close to the surrounding retinal tissue, making them difficult to segment reliably.

**Presence of normal anatomical structures:** The optic disc — a bright circular region where the optic nerve exits the eye — produces high-intensity pixels that can be confused with exudates. The retinal vasculature, a network of branching dark vessels, creates complex background structure that complicates lesion detection.

**Dataset variability:** Images collected from different clinical sites, cameras, and patient demographics vary substantially in brightness, contrast, field of view, and artifact presence (e.g., eyelashes, lens reflections). This makes models trained on one dataset difficult to generalize to others.

These challenges motivate a careful preprocessing stage before any segmentation or classification is attempted.

## III. Image Preprocessing and Enhancement

### A. Channel Selection

Fundus images are captured in RGB color space. Research has consistently shown that the green channel provides the highest contrast between retinal structures (vessels, lesions) and the background compared to the red or blue channels [4]. This is because hemoglobin, which gives blood vessels and hemorrhages their dark appearance, absorbs green light most strongly. Hard exudates, which are bright yellow, also stand out prominently in the green channel.

Accordingly, most preprocessing pipelines begin by extracting the green channel as a grayscale input. Some works instead convert to the HSV or LAB color space and operate on the value or luminance channel, which can be more robust to color calibration differences across cameras [5].

### B. Contrast Limited Adaptive Histogram Equalization (CLAHE)

Global histogram equalization (HE) spreads out the intensity distribution of an image to improve contrast. However, applied globally, it can over-amplify noise in homogeneous regions and distort local structures — problematic in fundus images where pathological regions are small and localized.

CLAHE [6] is a spatially adaptive variant that divides the image into non-overlapping rectangular tiles and applies histogram equalization within each tile independently. To prevent noise amplification, a *clip limit* is applied: if any histogram bin exceeds the clip limit, the excess is distributed uniformly across all bins before equalization. This preserves local contrast while controlling amplification of flat regions.

CLAHE has become a near-universal preprocessing step in fundus image analysis pipelines. Setiawan et al. [5] demonstrated that CLAHE with a clip limit of 2.0 and tile size $8 \times 8$ consistently improves the visibility of microaneurysms and exudates across multiple DR datasets compared to both raw input and global HE.

### C. Noise Suppression

After contrast enhancement, residual high-frequency noise is typically suppressed using a Gaussian filter. A Gaussian kernel with a small standard deviation ($\sigma \approx 1.0$, kernel size $5 \times 5$) smooths pixel-level noise while largely preserving the edges of larger structures such as vessels and exudate boundaries. Median filtering is an alternative that better preserves edges but is computationally heavier for large images.

### D. Background Normalization

Some pipelines apply additional background normalization by subtracting a blurred version of the image (using a large Gaussian kernel, e.g., $\sigma = 30$–$50$ pixels) from the original. This effectively removes the low-frequency illumination gradient caused by the fundus camera optics, yielding a more uniformly illuminated image. This technique, sometimes called *shade correction* or *background subtraction*, is particularly useful for images with strong center-periphery brightness variation [7].

### E. Optic Disc Exclusion

The optic disc is a bright, roughly circular region that can be easily confused with hard exudates in thresholding-based segmentation. Several methods exist for optic disc detection and exclusion: morphological reconstruction [7], template matching, and deep learning-based segmentation. In classical pipelines, the optic disc is detected as the brightest connected region of sufficient area, localized, and then masked out before lesion segmentation proceeds.

## IV. Lesion Segmentation

### A. Thresholding-Based Approaches

Thresholding converts a grayscale image to a binary mask by classifying pixels above a certain intensity as foreground (lesion) and below as background. For hard exudate detection, where exudates are bright against a darker retinal background, thresholding is a natural starting point.

**Otsu's Method** [8] computes an optimal global threshold by maximizing the inter-class variance between the foreground and background intensity distributions. It is parameter-free and computationally efficient, making it widely used as a first-pass segmentation step. The optimal threshold $t^*$ is found by:

$$t^* = \arg\max_t \left[\omega_0(t)\,\omega_1(t)\,(\mu_0(t) - \mu_1(t))^2\right] \qquad (1)$$

where $\omega_0(t)$, $\omega_1(t)$ are the probabilities of the two classes and $\mu_0(t)$, $\mu_1(t)$ are their respective mean intensities at threshold $t$.

The limitation of global thresholding is that it performs poorly when illumination is non-uniform across the image. Adaptive thresholding, which computes a local threshold for each pixel based on the statistics of its neighborhood, can be more robust in such cases, though it introduces more parameters (neighborhood size, constant offset) that require tuning.

### B. Morphological Operations

Binary masks obtained from thresholding invariably contain false positives: small isolated noise blobs that are bright by chance, and broken or fragmented true lesion regions. Morphological operations are used to refine these masks.

**Erosion** shrinks foreground regions by removing boundary pixels. **Dilation** expands foreground regions. Combining them yields two important composite operations:

- **Opening** (erosion followed by dilation): removes small isolated noise blobs while roughly preserving the size of larger true lesion regions.
- **Closing** (dilation followed by erosion): fills small gaps and holes within segmented regions, useful when true lesions appear as broken fragments.

The choice of structuring element (shape and size) matters significantly. A disk-shaped structuring element with a radius matched to the expected microaneurysm size (approximately 5–15 pixels at typical resolutions) is commonly used for MA detection.

### C. Vessel Segmentation

Since retinal vessels are normal anatomical structures and not lesions, separating them from pathological regions is important for accurate lesion detection. Blood vessels appear as dark, elongated, branching structures.

**Matched filtering** convolves the image with a Gaussian-shaped kernel oriented at multiple angles to respond strongly to elongated vessel-like structures [4]. The multi-angle responses are combined (usually by taking the maximum) to produce a vessel likelihood map, which can then be thresholded to produce a vessel mask.

**Morphological top-hat transform** is another classical approach: applying a morphological opening with a large structuring element estimates the background, and subtracting it from the original image highlights thin, bright (or dark) structures that do not fit within the structuring element — i.e., vessels [7].

### D. Deep Learning-Based Segmentation

U-Net [9] is a fully convolutional encoder-decoder architecture originally developed for biomedical image segmentation. Its skip connections between encoder and decoder layers allow it to combine low-level spatial detail with high-level semantic information, enabling precise boundary localization even for small structures. U-Net and its variants have been extensively applied to retinal vessel segmentation and lesion segmentation, achieving state-of-the-art results on benchmarks such as DRIVE and STARE for vessels, and DIARETDB1 for lesions.

The trade-off compared to classical methods is that U-Net requires annotated training data (pixel-level segmentation masks), which are expensive and time-consuming to produce. For DR lesions, pixel-level annotations are particularly laborious given the small size and subtle appearance of microaneurysms.

## V. Classification Approaches

### A. Feature-Based Classical Classifiers

Early automated DR systems extracted handcrafted features from segmented lesion regions and fed them to classical machine learning classifiers. Commonly used features include:

- **Morphological features:** Area, perimeter, compactness, and eccentricity of candidate lesion blobs.
- **Intensity features:** Mean, variance, and histogram statistics of pixel intensities within lesion regions.
- **Texture features:** Gray Level Co-occurrence Matrix (GLCM) descriptors (contrast, energy, correlation, homogeneity) computed within lesion or global retinal regions.
- **Spatial features:** Location of lesions relative to the optic disc and fovea.

These features were fed to classifiers such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Random Forests. Sinthanayothin et al. [10] used recursive region growing with a neural network classifier for exudate detection, achieving sensitivity of 80.21% and specificity of 70.66% on a small dataset. While interpretable and computationally lightweight, these approaches are sensitive to the quality of prior segmentation and generalize poorly across datasets with different imaging conditions.

### B. Convolutional Neural Network Classifiers

The introduction of large-scale labeled datasets and GPU-accelerated training enabled deep convolutional neural networks to dramatically outperform handcrafted feature approaches. Rather than manually designing features, CNNs learn hierarchical representations directly from image data through successive convolutional, pooling, and nonlinear activation layers.

Gulshan et al. [11] trained a deep CNN on a dataset of 128,175 retinal images with grades from ophthalmologists and achieved an AUC of 0.991 for detecting referable DR, with sensitivity of 97.5% and specificity of 93.4% — exceeding the performance of general ophthalmologists on the same task. This landmark study demonstrated the clinical viability of deep learning for DR screening.

Transfer learning has made CNN-based DR detection accessible at smaller scales. Architectures pretrained on ImageNet — such as VGG-16, ResNet-50, and EfficientNet — can be fine-tuned on DR datasets of a few thousand images, achieving strong performance without requiring millions of labeled fundus images. Gargeya and Leng [12] fine-tuned a

ResNet on 75,137 images and achieved an AUC of 0.97 for binary DR classification.

### C. Multi-Task and Attention-Based Approaches

More recent work explores multi-task learning (simultaneously predicting DR grade and localizing lesions) and attention mechanisms that guide the network to focus on diagnostically relevant regions. Attention maps can be overlaid on fundus images to provide clinician-interpretable explanations of model decisions, addressing a major limitation of black-box CNN classifiers. Porwal et al. [13] provide a benchmark for both lesion segmentation and DR grading on the IDRiD dataset, which is India-specific and therefore particularly relevant for domestic healthcare applications.

## VI. Comparative Summary of Reported Results

Table I summarizes the performance of representative methods from the literature across commonly used datasets. All figures are taken directly from the cited papers.

TABLE I
Reported Performance of DR Detection Methods in Literature

| Method | Dataset | Acc. | Sens. | AUC |
|---|---|---|---|---|
| Sinthanayothin et al. [10] (SVM + handcrafted) | Private (89 img) | — | 80.2% | — |
| Gulshan et al. [11] (Deep CNN) | EyePACS + Messidor | — | 97.5% | 0.991 |
| Gargeya & Leng [12] (ResNet) | Kaggle DR | — | 93.0% | 0.970 |
| Porwal et al. [13] (U-Net variants) | IDRiD | — | — | 0.901 |
| Setiawan et al. [5] (CLAHE + SVM) | Messidor | 88.0% | 86.0% | — |

The table highlights a clear progression: classical feature-based methods operating on small, controlled datasets give way to deep learning approaches trained on large-scale data, with correspondingly large gains in sensitivity. However, deep learning results from large proprietary datasets (e.g., Gulshan et al., using Google's internal data) are difficult to reproduce in academic settings. Transfer learning on public datasets (Kaggle DR, IDRiD, Messidor, APTOS 2019) represents the more practically accessible approach for researchers.

## VII. Proposed Pipeline Design

Based on the surveyed literature, we outline a principled pipeline design for a DR detection system, integrating the most well-supported techniques at each stage. This constitutes the proposed contribution of this work. The pipeline is illustrated conceptually in Fig. 1.

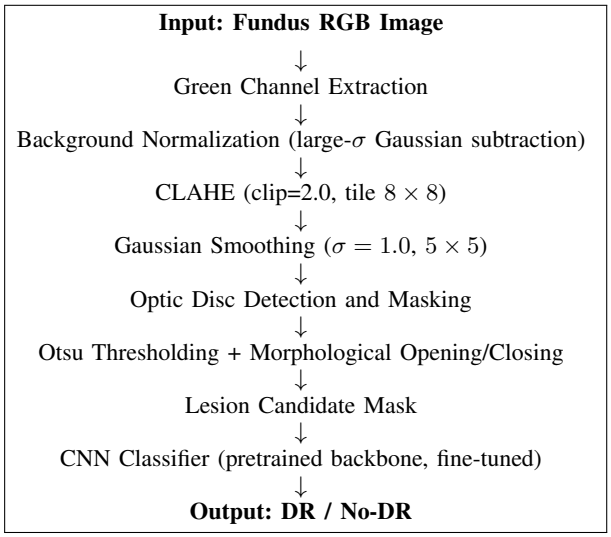



Fig. 1. Proposed end-to-end DR detection pipeline.

**Design Rationale:** Green channel extraction is chosen based on its consistently reported superiority for retinal structure contrast [4]. Background normalization is included to handle the center-periphery illumination gradient before CLAHE, preventing CLAHE from wasting its dynamic range on illumination rather than pathology. Optic disc masking is included as a necessary step before thresholding, since the optic disc is the most common source of bright false positives in exudate detection [7]. Morphological opening with a disk structuring element of radius 2 is chosen to remove noise blobs smaller than the smallest clinically significant lesion (approximately 5 pixels at standard resolution), while closing fills fragmented detections.

For the classification stage, we recommend a pretrained ResNet-50 or EfficientNet-B0 backbone fine-tuned on APTOS 2019 [14], as these achieve strong performance with datasets of a few thousand images. The preprocessed and masked image — rather than the raw RGB — should be used as the CNN input, so that the network benefits from the contrast normalization and lesion emphasis introduced by the classical pipeline.

## VIII. Discussion

### A. Classical vs. Deep Learning Approaches

The surveyed literature makes clear that deep learning substantially outperforms classical feature-based methods when large labeled datasets are available. However, classical methods retain important advantages: they are interpretable (each step has a clear geometric or statistical meaning), computationally inexpensive (suitable for deployment on low-power hardware), and robust when data is scarce.

A hybrid approach — using classical DIP for preprocessing and coarse segmentation, followed by a lightweight CNN for classification — strikes a reasonable balance. The classical stage reduces the complexity of the problem for the CNN (by normalizing inputs and highlighting lesion regions), potentially reducing the amount of training data needed.

### B. Dataset Considerations

Most high-performing results in the literature come from large, well-curated datasets collected in high-income countries. For deployment in India, datasets such as IDRiD (Kolhapur) [13] and APTOS 2019 (Aravind Eye Hospital, Tamil Nadu) [14] are more representative of the target population, imaging conditions, and camera hardware. Using such datasets for both training and evaluation is important for clinically relevant performance estimates.

### C. Evaluation Metrics

For a medical screening task, sensitivity (recall) is arguably the most important metric: missing a DR case (false negative) carries a much higher clinical cost than a false alarm that is subsequently cleared by an ophthalmologist. AUC-ROC provides a threshold-independent view of classifier performance and is the standard metric for comparing DR detection systems. Accuracy alone is insufficient given the potential class imbalance between No-DR and DR populations in screening cohorts.

### D. Remaining Challenges

Despite significant progress, several challenges remain open. Multi-class severity grading (distinguishing mild, moderate, severe, and proliferative DR) is substantially harder than binary detection and remains an active research problem. Cross-dataset generalization is still limited: models trained on one fundus camera system often degrade when tested on images from different cameras. Explainability — providing clinicians with visual evidence for a model's decision — is a practical requirement for clinical deployment but is not yet standardized. Finally, the detection of microaneurysms, the earliest DR sign, remains technically difficult due to their small size and low contrast.

## IX. Conclusion

This paper surveyed the landscape of image processing and machine learning techniques applied to automated diabetic retinopathy detection from retinal fundus photographs. We reviewed preprocessing methods (green channel extraction, CLAHE, Gaussian filtering, background normalization), segmentation approaches (Otsu thresholding, morphological operations, matched filtering, U-Net), and classification strategies (handcrafted features with classical classifiers, deep CNNs, transfer learning). A comparative summary of reported performance across benchmark datasets was presented.

Based on this analysis, we proposed a principled pipeline design combining a structured classical preprocessing stage with a fine-tuned CNN classifier. The design choices at each stage were justified with reference to the surveyed literature. Key remaining challenges include cross-dataset generalization, microaneurysm detection, and explainability for clinical acceptance.

Automated DR screening has the potential to make early detection accessible to large diabetic populations in resource-limited settings. Continued progress on publicly available, demographically diverse datasets — particularly India-specific ones — is essential for this potential to be realized clinically.


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